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Paper currency harbours antibiotic-resistant coliform bacteria and integron integrase

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Keywords

antimicrobial resistance, class 1 integrase gene, class 2 integrase gene, coliform bacteria, Gram negative bacteria, integrons, multiple drug resistance, paper currency.

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Abstract

Aims: This study was designed to analyse the prevalence of class 1 and class 2 integron integrase genes among antibiotic-resistant coliform bacteria isolated from paper currency circulating in Pakistan.

Methods and Results: A total of 500 individual currency notes were collected from different food vending sites at Lahore, Pakistan. Bacterial population were identified by biochemical and PCR techniques. Antimicrobial susceptibility testing was performed by disc diffusion assay. The highest bacterial population on currency was found from street vendors and butcher shops. *Escherichia coli* was found to be the most prevalent coliform bacteria followed by *Klebsiella* sp. and *Enterobacter* sp. PCR amplification of antimicrobial resistance gene showed the presence of *ampC*, *bla_{TEM}*, *bla_{NDM-1}*, *qnrA*, *tet(A)* and *tet(B)* genes among coliform isolates. A total of 47 integron integrase bearing strains of coliform bacteria were analysed. Sequence analysis showed the presence of *dfrA1-aadA1*, *dfrA1*, *dfrA5*, *dfrA7*, *aadA1*, *aadA4* cassette arrays in class 1 integron and *dfrA1-sat2-aadA1* in class 2 integrase genes.

Conclusion: Circulating currency was heavily contaminated with antimicrobial-resistant coliform bacteria bearing class 1 and class 2 integron integrase genes.

Significance and Impact of the Study: This study describes a potential threat of severe bacterial infections due to improper hand hygiene and community sanitation when dealing with the currency notes.

Introduction

For the first time paper money was introduced by China around 1000 AD (Tulloch 1957; Ireland 2003). Now the paper money has become widely exchanged for goods and other services worldwide. Currency note contamination came under spotlight in late 1880s and early 1990s, when scientists hypothesized the transmission of micro-organisms through currency (Schaarschmidt 1884). Paper currency could serve as fomite based on the ideal habitat for the growth of micro-organisms due to many advantageous features. First, they provide large surface area and organic debris to grow; second, they provide settling site for micro-organisms due to folds, depression and projections on the paper currency notes; third, the currency notes weave their way different populations over the years; lastly the currency notes and coins packed in leather, cotton and polythene bags usually stored in dark

and moist environment which favours the growth of micro-organisms (Lamichhane *et al.* 2009; Ayandele and Adeniyi 2011; Pradeep *et al.* 2012).

People living in unsanitary environments and having unhealthy practices will infect the currency notes with microbes, for example, counting the currency notes using saliva, sneezing and coughing on hands before the exchange of money and lack of hand washing habit leads to contamination and these currency notes will act as a source of vehicle carrying microbes to contaminate the hands of the next user (Barolia *et al.* 2011). Particularly for the food outlet which heavily depended upon exchange of currency for their food like ready-to-eat products, the risk of cross-contamination increases.

Adherence of bacteria and other micro-organisms to the surfaces like paper money depends upon the factors that contain surface charge, surface roughness, stiffness and hydrophobic properties (Blanco *et al.* 1991). Moreover,

bacteria have huge potential to persist in highly adverse environments. Their affinity to attach on different surfaces and to procedure biofilms are the most important approaches for existence in diverse environment (Heyba *et al.* 2015). Many biofilm-forming bacteria including some pathogenic drug-resistant micro-organisms are also associated with paper currency (Lamichhane *et al.* 2009).

The growing concern of antibiotic-resistant bacteria has turned attention towards the role of environment. Recently, the role of environment en route to antibiotic resistance has been acknowledged (Bengtsson-Palme *et al.* 2017). Integrons are common and an ancient feature of bacteria and hotspot of genotypic diversity, genome complexity and spread of antibiotic-resistant genes (Gillings 2014). Class 1 integron integrase gene has been considered as a genetic marker of anthropogenic pollutants (Gillings *et al.* 2015). Because integrons are linked to the genes mediating antibiotics and heavy metal resistance (Partridge *et al.* 2001), have penetration power into antibiotic-resistant pathogens, commensal bacteria of humans (Stokes and Gillings 2011) and frequent transferability between bacteria due to the location on mobile genetic element.

This study was designed to investigate the prevalence and antibiotic resistance pattern of coliforms in the circulating currency. Keeping in view the unique characters of integrase gene its insertion into diverse genetic elements, penetration into bacterial species associated with humans and animal infections, carriage and transportation of antibiotic-resistant genes, wide geographic distribution and rapid increase in abundance. Class 1 and class 2 integron integrase genes were also analysed as a proxy for anthropogenic pollution among coliform isolates. This study will be helpful in identifying the hidden source of microbial communities and identifying phenotypic and genotypic basis of antimicrobial resistance in coliforms. Therefore, awareness will be generated in the management practices particularly personal hygiene and hand washing.

Materials and methods

Collection of currency notes

Different currency notes were randomly collected with lower denomination (worth Rs. 10, Rs. 20, Rs. 50 and Rs. 100) and higher denomination (worth Rs. 500, Rs. 1000 and Rs. 5000). These currency notes were collected from food vending sites categorized as butcher shops, bakery and confectionery shops, fast food outlets, coffee shops and street vendors at Lahore, Pakistan. A total of 500 individual currency notes were collected, separately. In addition, 50 new paper currency notes obtained from bank (Muslim Commercial Bank Ltd., Lahore, Pakistan) as negative control. All collected samples were free from

any dirt, stains or visible contamination signs. Samples were collected using sterile techniques in sterile zip-lock bags and carefully transported to Microbiology laboratory at University of Central Punjab, Lahore, Pakistan.

Isolation of bacteria from collected currency notes

Sterile cotton swabs dipped in autoclaved tryptic soy broth (TSB; Difco, Detroit, Mich.) was carefully surface swabbed covering all parts of currency notes including front and back sides. Serially diluted 100 µl TSB from each sample was streaked on MacConkey agar (Oxoid, Basingstok, UK) and Eosine Methylene Blue (EMB) agar (Sigma-Aldrich, St. Louis, MO) plates and incubated at 37°C for 24 h, separately. Pink and black coloured colonies with green sheen were isolated from MacConkey and EMB agar plates, respectively. Lactose fermenter bacterial strains were further sub-cultured on MacConkey agar to obtain pure colonies.

Biochemical and molecular characterization of bacterial isolates

Bacterial isolates were biochemically characterized using API-20E kit (BioMérieux, Marcy-l'Etoile, France) (Shayegani *et al.* 1978). After biochemical characterization, bacterial isolates were grown overnight in 10 ml LB broth. Broth culture was centrifuged at 10 000 g, supernatant was discarded and cell pellets were used for DNA extraction. DNA extraction was performed using commercially available GeneJET Genomic DNA extraction kit (Thermo Fischer Scientific, Waltham, MA). Using DNA as template, molecular identification of the isolates was performed by polymerase chain reaction (PCR) in PCR tubes using 16S rRNA pA and pH primers described previously by Edwards *et al.* (1989). The reaction was carried out in 25 µl volume including 12.5 µl PCR master mix (Thermo Fischer Scientific), 1 µl DNA template, 1 µl each forward and reverse primers and 9.5 µl dH₂O. PCR reaction was set at 95°C as initial denaturation for 5 min followed by 35 PCR cycles consisting of denaturation step at 95°C for 30 s, annealing at 60°C for 1 min, extension at 72°C for 1 min and final extension at same temperature for 10 min.

Antimicrobial susceptibility testing

Bacterial strains were grown overnight in 10 ml LB broth and 100 ml each bacterial suspension was spread on Muller Hinton Agar (Oxoid) plates. Commercially available antibiotic disks were used to check the antibacterial activity against bacterial strains. Antimicrobial susceptibility testing was performed by Clinical and Laboratory Standards Institute guidelines (Clinical and Laboratory Standards Institute

2011). A panel of antibiotic discs were used as follows: amoxicillin (10 µg), tetracycline (30 µg), gentamycin (10 µg), streptomycin (10 µg), piperacillin (30 µg), cefotaxime (30 µg), ciprofloxacin (5 µg), meropenem (10 µg), trimethoprim/sulfamethoxazole (1.25 µg/ 23.75 µg).

PCR amplification of antimicrobial-resistant genes

Bacterial strains phenotypically exhibiting antimicrobial resistance against antibiotics, were subjected to PCR amplification with their respective antimicrobial resistance genes. PCR amplification was performed for beta-lactam antibiotics resistance genes including *ampC* (Pérez-Pérez and Hanson 2002), *bla_{TEM}* (Bailey et al. 2010), *bla_{NDM-1}* (Xiang et al. 2020) and *bla_{CTX-M}* (Pagani et al. 2003). Quinolones resistance genes *qnrA* and *qnrS* were amplified by following the protocol as mentioned by Lavilla et al. (2007). Similarly, multiplex PCR was designed for the amplification of tetracycline resistance genes including *tet(A)*, *tet(B)* and *tet(C)* (Ng et al. 2001).

Gene amplification and genetic analysis of Class 1 and 2 integrase gene

Genomic DNA from selected bacterial strains were used to check the prevalence of class 1 and 2 integrase

gene (Table 1) by multiplex PCR as described by Kheiri and Akhtari (2016). PCR reaction was carried out in 25 µl PCR tubes as described above. PCR reaction was adjusted as initial denaturation at 95°C for 5 min followed by 30 cycles of denaturation at 95°C for 1 min, annealing at 59°C for 1 min, extension at 72°C for 1 min and final extension at same temperature for 8 min. PCR amplification of variable regions in class 1 and 2 integrase gene was performed using primers complementary to 5' and 3' conserved region of class 1 (Levesque et al. 1995) and class 2 (White et al. 2001) integrase genes (Table 1). The amplified product was purified using PCR gel purification kit (QIAquick; Qiagen, Valencia, CA). Purified product was sent for sequencing to Macrogen Inc. Korea. Nucleotide sequences were aligned with GenBank data by using BLAST tool from NCBI. Those gene cassettes were selected and identified whose similarity with GenBank data was at least 95%.

Statistical analysis

The results were collected in triplicates ($n = 3$) as mean $\pm$ SD. All variables were subjected to analysis of variance. Compared means were separated by Duncan's multiple range test at a significant level ($P < 0.05$).

Table 1 Primers used in this study

Target gene	Sequence (5'–3')	Reference
<i>bla_{TEM}</i> (RH605)	TTTCGTGTCGCCCTTATTCC	Bailey et al. (2010)
<i>bla_{TEM}</i> (RH606)	CCGGCTCCAGATTATCA GC	
<i>bla_{NDM-1}</i> -F	GGCGGAAGGCTCATCACGA	Xiang et al. (2020)
<i>bla_{NDM-1}</i> -R	CGCAACACAGCCTGACTTTC	
<i>bla_{CTX-M}</i> U1	ATGTGCACCAGTAARGT	Pagani et al. (2003)
<i>bla_{CTX-M}</i> U2	TGGGTAAARTARGTSACCAGA	
<i>ampC</i> -F	AACACACTGATTGCGTCTGAC	Pérez-Pérez and Hanson (2002)
<i>ampC</i> -E	CTGGGCCTCATCGTCAGTTA	
<i>tet(A)</i> -F	GCTACATCCTGCTTGCCCTTC	Ng et al. (2001)
<i>tet(A)</i> -R	CATAGATCGCCGTGAAGAGG	
<i>tet(B)</i> -F	TTGGTTAGGGGCAAGTTTTG	Ng et al. (2001)
<i>tet(B)</i> -R	GTAATGGGCCAATAACACCG	
<i>tet(C)</i> -F	CTTGAGAGCCTCAACCCAG	Ng et al. (2001)
<i>tet(C)</i> -R	ATGGTCGTCATCTACCTGCC	
<i>qnrA</i> (qnrFamF)	ACGCCAGGATTGAGTGAC	Lavilla et al. (2007)
<i>qnrA</i> (qnrFamR)	CCAGGCACAGATCTTGAC	
<i>qnrS</i> -F	CCTACAATCATACATATCGGC	Lavilla et al. (2007)
<i>qnrS</i> -R	GCTTCGAGAATCAGTTCTTGC	
Integrase class I (<i>IntI</i> . F)	TCTCGGGTAACATCAAGG	Kheiri and Akhtari (2016)
Integrase class I (<i>IntI</i> . R)	GTTCTTCTACGGCAAGGT	
Integrase class II (<i>IntII</i> . F)	CACGGATATGCGACAAAAAGGT	Kheiri and Akhtari (2016)
Integrase class II (<i>IntII</i> . R)	GTAGCAAACGAGTGACGAAATG	
Class I integrase variable Region (F)	GGCATCCAAGCAAG	Levesque et al. (1995)
Class I integrase variable Region (R)	AAGCAGACTTGACCTGA	
Class II integrase variable Region (F)	GATGCCATCGCAAGTACGAG	White et al. (2001)
Class II integrase variable Region (R)	CGGGATCCCGGACGGCATGCACGATTGTGA	

Statistical analysis was performed using IBM SPSS software (IBM, SPSS Statistics, ver. 25.0).

Results

The number of bacteria on currency notes obtained from different food outlets varied greatly. Total 500 currency notes were screened for the presence of micro-organisms. Four hundred sixty-eight currency notes out of 500 were contaminated with coliform bacteria (93.6%). While no coliform bacteria were isolated from 50 new currency notes collected from banks. From the samples, total 220 bacterial strains were isolated and out of which 120 bacterial strains were belonged to the coliform group comprises seven major coliform bacteria (Fig. 1) including *E. coli* (61%), *Klebsiella pneumoniae* (26%), *Enterobacter cloacae* (8%) and *Citrobacter freundii* (3%), *Proteus mirabilis* (2%), *Shigella* sp. (0.8%) and *Salmonella typhi* (0.45%).

Bacterial numbers on lower and higher currency denominator

Coliform bacterial population was determined by number of bacteria per square cm (cm²) on each currency note. It was also observed that lower currency denominator for example, Rs. 10 to Rs. 100 were more contaminated with microbes as compared to higher currency denominator (Fig. 1) like Rs. 500, Rs. 1000 and Rs. 5000 ($P < 0.05$). Rs. 10 currency notes were contaminated with *E. coli* (68%), *K. pneumoniae* (16%) and *E. cloacae* (16%). Rs. 20 currency notes were contaminated with *E. coli* (56%), *K. pneumoniae* (36%), *E. cloacae* and *C. freundii* (4%). Rs. 50 notes in which 64% were *E. coli*, 28% *K. pneumoniae*, 4% were prevalent. Rs. 100 currency notes were found to contain 60% *E. coli*, 32% *K. pneumoniae*, 4% *E. cloacae* and *C. freundii*, respectively. Rs. 500 currency notes were contaminated with *E. coli* (62.5%), *K.*

pneumoniae (29%), *E. cloacae* (4.3%) and *C. freundii* (4.3%). Rs. 1000 and Rs. 5000 currency notes were contaminated with *E. coli*, *Klebsiella* and *Enterobacter* with 77.3, 18.2, 4.5% and 59.1, 22.7, 13.7 and 4.5% respectively (Fig. 1).

Bacterial number in different food vending outlets

Currency notes collected from different food vending sites including butcher shops, bakery & confectionery shops, fast food restaurants, coffee shops and street vendors exhibited contamination of coliform bacteria. However, the number of coliform bacteria varied among different food points. For example, currency notes collected from butcher shops were contaminated with *E. coli* (69%), *K. pneumoniae* (23%), *E. cloacae* (5.5%) and *C. freundii* (2.5%). Specimens from bakery and confectionery shops were contaminated with *E. coli* (63.5%), *K. pneumoniae* (28.5%), *E. cloacae* (5.5%) and (2.5%) *C. freundii*. Fast food restaurants were found to contain *E. coli* (57.2%), *K. pneumoniae* (25.7%), *E. cloacae* (11.4%) and *C. freundii* (5.7%). Currency from street vendors contained *E. coli* (66.6%), *K. pneumoniae* (24.4%) and *E. cloacae* (9%). Similarly, currency collected from coffee shops contained *E. coli* (63.4%), *K. pneumoniae* (30%), *E. cloacae* (3.3%) and *C. freundii* (3.3%). The highest CFU per cm² was observed among currency notes collected from butcher shop and street vendors (Fig. 2a,b). Whereas, currency notes collected from coffee shop and bakery and confectionery shops demonstrated lowest CFU per cm² ($P < 0.05$) (Fig. 2a,b).

Bacterial numbers influenced by currency age

Pakistani currency notes are embedded with the production year. Hence, it made easy to estimate an age of the currency notes. We collected up to 5 years old currency notes (i.e. from year 2015–2020). In our results, we found

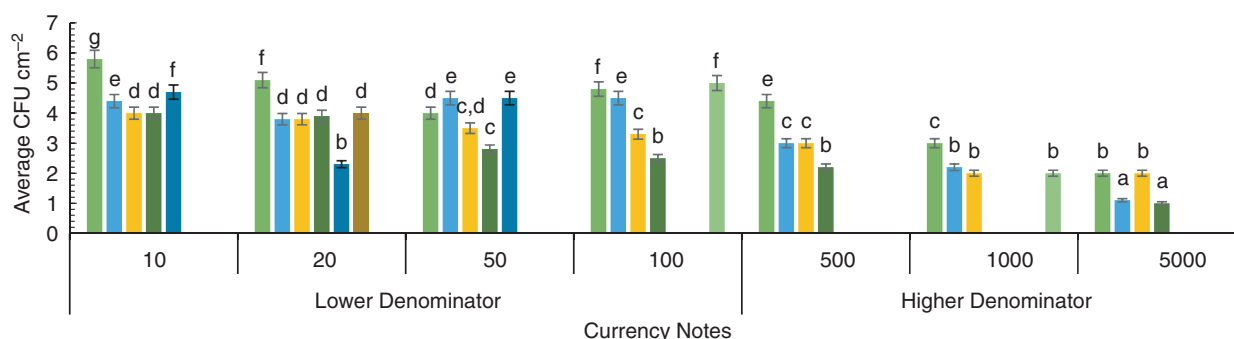


Figure 1 Number of bacteria (CFU per cm²) on lower and higher denominator currency notes. Error bars representing mean $\pm$ SE of triplicates. The alphabetical letters above the bars indicate significant differences between treatments calculated by Duncan's Multiple Range Test (DMRT) (*E. coli*; *Klebsiella*; *Enterobacter*; *Citrobacter*; *Proteus*; *Shigella*; *Salmonella*). [Colour figure can be viewed at wileyonlinelibrary.com]

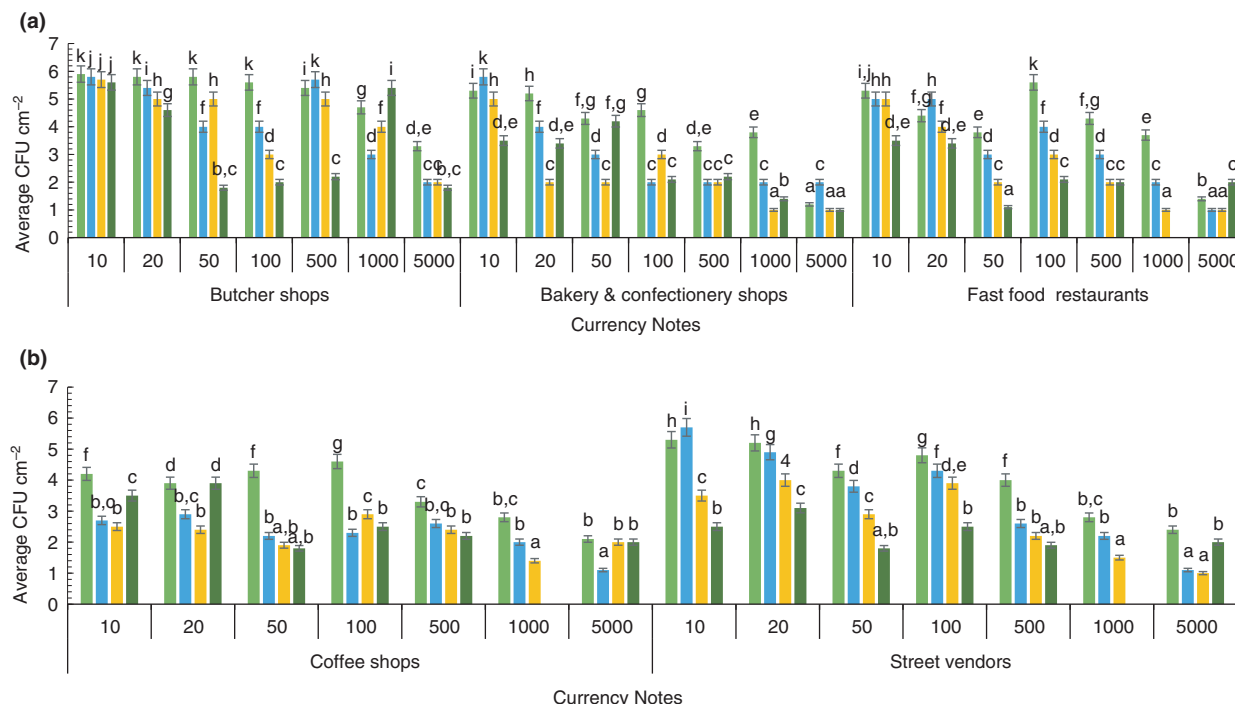


Figure 2 (a) Number of bacteria (CFU per cm^2) on currency notes collected from different food vending outlets. Error bars representing mean $\pm$ SE of triplicates. The alphabetical letters above the bars indicate significant differences between treatments calculated by Duncan's multiple range test (DMRT). (b) Number of bacteria (CFU per cm^2) on currency notes collected from different food vending outlets. Error bars representing mean $\pm$ SE of triplicates. The alphabetical letters above the bars indicate significant differences between treatments calculated by Duncan's Multiple Range Test (DMRT) (■ *Escherichia coli*; ■ *Klebsiella*; ■ *Enterobacter*; ■ *Citrobacter*). [Colour figure can be viewed at wileyonlinelibrary.com]

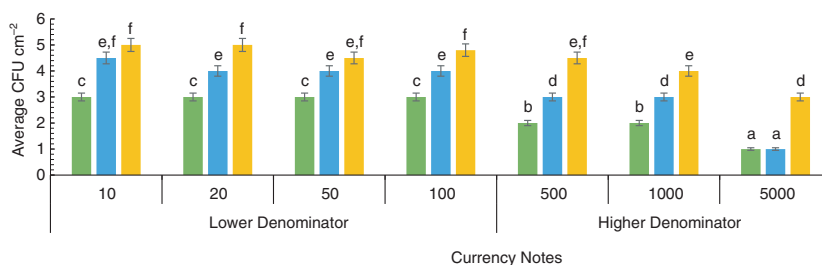


Figure 3 Influence of age (years) from 2015 to 2020 on the bacterial content (CFU per cm^2) of currency notes ($P < 0.05$). Error bars representing mean $\pm$ SE of triplicates. The alphabetical letters above the bars indicate significant differences between treatments calculated by Duncan's Multiple Range Test (DMRT) (■ 2019–2020; ■ 2017–2018; ■ 2015–2016). [Colour figure can be viewed at wileyonlinelibrary.com]

a linear correlation between the age of the notes (in years) and bacterial load. Number of bacteria approximately increased one logarithmic unit after 2 years ($P < 0.05$) and highest number of bacteria (CFU per cm^2) was observed among currency notes having production year 2015–2016 (Fig. 3).

Antibiotic susceptibility testing

Of the 120 coliform bacteria, 38 (31.6%) were multidrug-resistant against two three or more antibiotics. 52

(43.3%) were resistant to one antibiotic and 17 (25%) were sensitive to all tested antibiotics. *Escherichia coli* represented 31.5% antibiotic-resistant isolates. *Klebsiella* was second most prevalent genera with 38.7% antibiotic-resistant isolates ($P < 0.05$). None of the bacteria identified as *Salmonella* exhibited bacterial resistance. In contrast, all *Proteus* isolates were resistant to at least one antibiotic. Exceptionally high resistance was observed against beta-lactam amoxicillin antibiotic. Thirty-three (27.5%) bacterial strains were found resistant to beta-lactam antibiotics including 18 isolates (54.5%) carried *ampC* gene, 10

Table 2 Occurrence of antibiotic-resistant genes and integrase gene in antibiotic-resistant bacterial isolates

	<i>Escherichia coli</i>	<i>Klebsiella</i> sp.	<i>Enterobacter</i> sp.	<i>Citrobacter</i> sp.	<i>Proteus</i> sp.	<i>Shigella</i> sp.	<i>Salmonella</i> sp.
Beta lactam antibiotics resistance (<i>n</i> = 33)							
<i>ampC</i>	33.4%	28.7%	15.6%	18.9%	2.1%	1.3%	—
<i>bla_{TEM}</i>	49.7%	32.5%	—	—	17.7%	—	—
<i>bla_{NDM-1}</i>	12.5%	6.2%	—	—	1.6%	—	—
<i>bla_{CTX-M}</i>	—	—	—	—	—	—	—
Tetracycline resistance (<i>n</i> = 9)							
<i>tet(A)</i>	9.7%	—	6.8%	—	2.9%	—	—
<i>tet(B)</i>	6.3%	—	—	—	4.1%	—	—
<i>tet(C)</i>	—	—	—	—	—	—	—
Quinolones resistance (<i>n</i> = 8)							
<i>qnrA</i>	40%	60%	—	—	—	—	—
<i>qnrS</i>	—	—	—	—	—	—	—
Class 1 integrase (<i>intI1</i>) cassette array (<i>n</i> = 47)							
<i>dfrA1-aadA1</i>	33.9%	28.6%	—	—	19.4%	—	—
<i>dfrA1</i>	32.5%	—	—	—	6.4%	—	—
<i>dfrA5</i>	—	22.1%	—	—	2.1%	—	—
<i>dfrA7</i>	2.6%	—	—	1.3%	1.4%	—	—
<i>aadA1</i>	6.9%	7.4%	—	17.6%	—	—	—
<i>aadA4</i>	7.8%	1.5%	—	—	2.7%	—	—
Class 2 integrase (<i>intI2</i>) cassette array (<i>n</i> = 6)							
<i>dfrA1-sat2-aadA1</i>	60%	—	—	—	40%	—	—

(30.3%) isolates with *bla_{TEM}* and 5 (15%) found *bla_{NDM-1}* positive. Tetracycline resistance was observed by amplification of *tet(A)*, *tet(B)* and *tet(C)* genes in 9 (7.5%) tetracycline resistant isolates. Five isolates (55.5%) were detected with *tet(A)* gene and four isolates (45.5%) had *tet(B)* gene while *tet(C)* gene was not detected in any isolate. All tetracycline-resistant isolates were also resistant to ampicillin ($P > 0.05$). Resistance against quinolones (ciprofloxacin) was analysed by *qnrA* and *qnrS* genes. Eight bacterial isolates were found positive with *qnrA* gene. Four ampicillin-resistant bacterial isolates were also resistant to ciprofloxacin (Table 2).

Integrase gene

Integrase gene analysis was performed on the coliform bacteria isolated from currency notes. Bacterial strains harbouring integrase gene were resistant to most of the tested antimicrobial agents ($P < 0.05$). Of 120 coliform bacteria, 47 (39.1%) were positive for class 1 integrase gene (*intI1*) and 6 (5%) were positive for class 2 integrase gene (*intI2*). While no coliform bacteria were positive for both *intI1* and *intI2* gene. The gene cassette arrays of integrase gene were analysed by PCR amplification and sequencing their variable regions. Class 1 integrase variable region contained five different gene cassettes arrays with size ranged from 0.18 to 1.9 kbp. Among 47 class 1 integrase positive isolates, 21 (44.6%) isolates carried *dfrA1-aadA1* gene cassette array, followed by *dfrA1*, *dfrA5*, *dfrA7* *aadA1* and *aadA4* gene cassettes in 12

(25.5%), 7 (14.8%), 4 (8.5), 2 (4.2%) and 1 (2.12%) isolates. Of the 6 class 2 positive integrase isolates, 4 (66.6%) isolates contained the *dfrA1-sat2-aadA1* gene cassette array with molecular size of 2.2 kbp. Gene cassette array in remaining two samples of class 2 integrase positive isolates could not be detected by PCR (Table 2).

Discussion

The analysis of coliform bacteria isolated from currency notes denotes the prevalence of antimicrobial resistance in various coliform isolates. This could be explained by the misuse and overuse of antibiotics prescribed for humans and animals. The overuse of antibiotics can be narrated by its annual production. For example, amoxicillin annual production in 2005 was 80 tons which is equivalent to 80 million daily doses annually (Kresken 2009). Not only amoxicillin, other antibiotics including sulphonamide, tetracycline, beta-lactams and trimethoprim are extensively produced and used to treat human and animal bacterial infection (Watkinson *et al.* 2009; Gao *et al.* 2012). Therefore, in the modern era, the detection of antibiotic-resistant coliform bacteria from currency note is not surprising (Rosas *et al.* 2015; Zhang *et al.* 2015).

In this study, currency notes were collected and analysed from five different groups including butcher shops, bakery and confectionery shops, fast food outlets, coffee shops and street vendors. The currency notes and coins specimens analysed for morphological, biochemical and antimicrobial resistant and also daily use of notes and

coins. It was observed that currency specimens collected from butcher shops and street vendors were heavily contaminated with the coliform bacteria as compared to the currency specimens collected from bakery and confectionery shops, fast food outlets and coffee shops. This behaviour could be explained by the overall hygiene practices among personals working in these areas. Particularly, washing and cleaning of hands before and after the use of currency notes and coins. Socio-economic status of the money and food handler largely influence on the disparity in the number of bacteria found on currency notes.

Moreover the constituents of currency notes and environmental conditions also effects the microbial population on currency notes. Usually paper currency notes are manufactured from cotton and mixed with other textile fibres like linen and abaca (Gedik *et al.* 2013). Due to humid and hot climatic conditions of Pakistan, currency notes became hotspot of bacterial populations. Most of them harbouring antimicrobial resistance genes which can cause serious life-threatening infections. This scenario can be easily explained by the current outbreak of COVID-19; in which paper currency can be responsible for the spread of SARS-CoV-2 (Kakushadze and Liew 2020).

Pakistan is a developing country with poor sanitary conditions. Enteric infections account almost 1.7 million lives globally in 2017 and majority of such cases were reported from low income countries (Roth *et al.* 2018). Paper currency and coins are common exchangeable medium for fomites and can easily transmit pathogenic micro-organisms (Angelakis *et al.* 2014). Current study was designed with the emphasis to coliform bacteria from currency notes and coins because it was hypothesized that currency notes would be involved in a route of faecal-oral infections. We observed variety of coliform bacteria including *E. coli* (61%), *Klebsiella pneumoniae* (26%), *Enterobacter cloacae* (8%) and other organisms were in less number. Similar research was conducted by Pereira da Fonseca *et al.* (2015) with Brazilian currency notes and found *Salmonella*, *Staphylococcus* and *Klebsiella* as abundant micro-organisms. In another study, Iranian currency was collected from different shops included food outlets, greengrocery, stores, pastry shops, sweet shops and poultry meat shops. Among other micro-organisms, *E. coli* was found most abundant which is correlated with the findings of our study (Moosavy *et al.* 2013). Vriesekoop *et al.* (2010) conducted a study on banknotes collected from 10 different countries. They found that *E. coli* is commonly isolated from the banknotes of China and USA; and *Salmonella* sp. was only isolated from China, USA and Ireland. Only few isolates of more severe form of coliform bacteria *Salmonella* sp. were isolated from

street vendors and could not found in any other sample. It is unclear that the presence of *Salmonella* in currency notes could be a source of current typhoid fever outbreak in Pakistan. However, the presence of such pathogenic micro-organism is indicative of poor hygiene practices or due to the conditions where such items are stored. Moreover, currency notes deterioration in terms of currency circulation time was calculated from its year of production. It was also observed that more the currency note deteriorated (years) more the bacterial load it had ($P < 0.05$). This could be another sign of currency handling and overall hygiene practices of a country.

Antimicrobial-resistant coliform bacteria with integrons integrase genes poses a serious alarming health issue because integron gene can transfer antimicrobial-resistant genes to other bacterial strains. Gene cassette arrays of class 1 and class 2 integron integrase genes confer resistance to several classes of antimicrobial agents. Mostly *dfrA*, *aadA* and *sat* genes cassette arrays frequently found in class 1 and class 2 integron respectively (Hall 2012). In this study, the most prevalent gene cassettes were *dfr* and *aad* which were in agreement with the previous study conducted by Kheiri and Akhtari (2016). *aad* gene cassette array family encodes aminoglycoside adenylyl transferases confers streptomycin/spectinomycin resistance (Partridge *et al.* 2009). A strong correlation was found in our study among isolates carrying *aad* cassette array and phenotypic resistance against streptomycin ($P < 0.05$). Other most common gene cassette array was *dfr* which encodes dihydrofolate reductases enzyme and cause trimethoprim/sulfamethoxazole (SXT) resistance.

In this research, we found that circulating currency is heavily contaminated with coliform bacteria. Simultaneous handling of food, and currency may spread a disease of public health concern. Due to the presence of integron integrase gene, transfer of antimicrobial resistance genes among bacteria became possible. This highlights the potential of currency notes and coins to spread nosocomial infections. To overcome this issue, three main strategies could be adopted, one is the change of cotton based currency notes to polymer/plastic based currency, secondly reducing the half-life of currency circulation which can lead to lower the bacterial load and thirdly improving overall hygiene practices which includes a standard hand washing practice particularly during food-currency exchange and educating the masses about the repercussion of handling contaminated currency.

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Conflict of Interest

The authors declared no conflict of interest.

References

- Angelakis, E., Azhar, E.I., Bibi, F., Yasir, M., Al-Ghamdi, A.K., Ashshi, A.M., Elshemi, A.G. and Raoult, D. (2014) Paper money and coins as potential vectors of transmissible disease. *Future Microbiol* **9**, 249–261.
- Ayandele, A. and Adeniyi, S. (2011) Prevalence and antimicrobial resistance pattern of microorganisms isolated from Naira notes in Ogbomoso North, Nigeria. *J Res Bio* **8**, 587–593.
- Bailey, J.K., Pinyon, J.L., Anantham, S. and Hall, R.M. (2010) Commensal *Escherichia coli* of healthy humans: a reservoir for antibiotic-resistance determinants. *J Med Microbiol* **59**, 1331–1339.
- Barolia, S.K., Verma, S. and Verma, B.K. (2011) Coliform contamination on different paper currency in Ajmer, Rajasthan, India. *Univers J Environ Res Technol* **1**, 552–556.
- Bengtsson-Palme, J., Kristiansson, E. and Larsson, D.J. (2017) Environmental factors influencing the development and spread of antibiotic resistance. *FEMS Microbiol Rev* **42**, 53.
- Blanco, J., Blanco, M., Garabal, J.I. and Gonzalez, E.A. (1991) Enterotoxins, colonization factors and serotypes of enterotoxigenic *Escherichia coli* from humans and animals. *Microbiologia* **7**, 57–73.
- Clinical and Laboratory Standards Institute (2011) *Performance Standards for Antimicrobial Susceptibility Testing*. Wayne, PA: Clinical and Laboratory Standards Institute.
- Edwards, U., Rogall, T., Blocker, H., Emde, M. and Bottger, E.C. (1989) Isolation and direct complete nucleotide determination of entire genes. Characterization of a gene coding for 16S ribosomal RNA. *Nucleic Acids Res* **17**, 7843–7853.
- Gao, P., Munir, M. and Xagoraki, I. (2012) Correlation of tetracycline and sulfonamide antibiotics with corresponding resistance genes and resistant bacteria in a conventional municipal wastewater treatment plant. *Sci Total Environ* **421**, 173–183.
- Gedik, H., Voss, T.A. and Voss, A. (2013) Money and transmission of bacteria. *Antimicrob Resist Infect Control* **2**, 22.
- Gillings, M.R. (2014) Integrons: past, present, and future. *Microbiol Mol Biol Rev* **78**, 257–277.
- Gillings, M.R., Gaze, W.H., Pruden, A., Smalla, K., Tiedje, J.M. and Zhu, Y.-G. (2015) Using the class 1 integron-integrase gene as a proxy for anthropogenic pollution. *ISME J* **9**, 1269–1279.
- Hall, R.M. (2012) Integrons and gene cassettes: hotspots of diversity in bacterial genomes. *Ann N Y Acad Sci* **1267**, 71–78.
- Heyba, M., Ismaiel, M., Alotaibi, A., Mahmoud, M., Baqer, H., Safar, A., Al-Sweih, N. and Al-Ta'ar, A. (2015) Microbiological contamination of mobile phones of clinicians in intensive care units and neonatal care units in public hospitals in Kuwait. *BMC Infect Dis* **15**, 434.
- Ireland, P. (2003) Bernholz, P. Monetary regimes and inflation: history, economic and political relationships. *Econ Polit Relationships* **80**, 272–276.
- Kakushadze, Z. and Liew, J.-K.-S. (2020) Coronavirus: case for digital money? *World Econ* **21**, 177–190.
- Kheiri, R. and Akhtari, L. (2016) Antimicrobial resistance and integron gene cassette arrays in commensal *Escherichia coli* from human and animal sources in IRI. *Gut Pathog* **8**.
- Kresken, M. (2009) GERMAP 2008-Resistance monitoring in Germany: antibiotic use and occurrence of resistance in human medicine. *J Vet Pharmacol Therapeut*, **32**, 51.
- Lamichhane, J., Adhikary, S., Gautam, P., Maharjan, R. and Dhakal, B. (2009) Risk of handling paper currency in circulation chances of potential bacterial transmittance. *Nepal J Sci Technol* **10**, 161–166.
- Lavilla, S., González-López, J.J., Sabaté, M., García-Fernández, A., Larrosa, M.N., Bartolomé, R.M., Carattoli, A. and Prats, G. (2007) Prevalence of qnr genes among extended-spectrum β -lactamase-producing enterobacterial isolates in Barcelona, Spain. *J Antimicrob Chemother* **61**, 291–295.
- Levesque, C., Piche, L., Larose, C. and Roy, P.H. (1995) PCR mapping of integrons reveals several novel combinations of resistance genes. *J Antimicrob Agents Chemother* **39**, 185–191.
- Moosavy, M.-H., Shavisi, N., Warriner, K. and Mostafavi, E. (2013) Bacterial contamination of Iranian paper currency. *Iran J Public Health* **42**, 1067–1070.
- Ng, L.K., Martin, I., Alfa, M. and Mulvey, M. (2001) Multiplex PCR for the detection of tetracycline resistant genes. *Mol Cell probe* **15**, 209–215.
- Pagani, L., Dell'Amico, E., Migliavacca, R., D'Andrea, M.M., Giacobone, E., Amicosante, G., Romero, E. and Rossolini, G.M. (2003) Multiple CTX-M-type extended-spectrum beta-lactamases in nosocomial isolates of Enterobacteriaceae from a hospital in northern Italy. *J Clin Microbiol* **41**, 4264–4269.
- Partridge, S.R., Brown, H.J., Stokes, H.W. and Hall, R.M. (2001) Transposons Tn1696 and Tn21 and their integrons In4 and In2 have independent origins. *Antimicrob Agents Chemother* **45**, 1263–1270.
- Partridge, S.R., Tsafnat, G., Coiera, E. and Iredell, J.R. (2009) Gene cassettes and cassette arrays in mobile resistance integrons. *FEMS Microbiol Rev* **33**, 757–784.
- Pereira da Fonseca, T.A., Pessôa, R. and Sanabani, S.S. (2015) Molecular analysis of bacterial microbiota on Brazilian currency note surfaces. *Int J Environ Res Public Health* **12**, 13276–13288.
- Pérez-Pérez, F.J. and Hanson, N.D. (2002) Detection of plasmid-mediated AmpC beta-lactamase genes in clinical

- isolates by using multiplex PCR. *J Clin Microbiol* **40**, 2153–2162.
- Pradeep, N., Marulasiddaiah, B. and Chetana, M. (2012) Microbial contamination of Indian currency notes in circulation. *J Res Biol* **2**, 377–382.
- Rosas, I., Salinas, E., Martínez, L., Cruz-Córdova, A., González-Pedrajo, B., Espinosa, N. and Amábile-Cuevas, C.F. (2015) Characterization of *Escherichia coli* isolates from an urban lake receiving water from a wastewater treatment plant in Mexico city: fecal pollution and antibiotic resistance. *Curr Microbiol* **71**, 490–495.
- Roth, G.A., Abate, D., Abate, K.H., Abay, S.M., Abbafati, C., Abbasi, N., Abbastabar, H., Abd-Allah, F. *et al.* (2018) Global, regional, and national age-sex-specific mortality for 282 causes of death in 195 countries and territories, 1980–2017: a systematic analysis for the Global Burden of Disease Study 2017. *Lancet* **392**, 1736–1788.
- Schaarschmidt, J. (1884) Upon the occurrence of bacteria and minute algae on the surface of paper money. *Nature* **30**, 360.
- Shayegani, M., Maupin, P.S. and McGlynn, D.M. (1978) Evaluation of the API 20E system for identification of nonfermentative Gram-negative bacteria. *J Clin Microbiol* **7**, 539–545.
- Stokes, H.W. and Gillings, M.R. (2011) Gene flow, mobile genetic elements and the recruitment of antibiotic resistance genes into Gram-negative pathogens. *FEMS Microbiol Rev* **35**, 790–819.
- Tullock, G. (1957) Paper money-a cycle in Cathay. *Econ History Rev* **9**, 393–407.
- Vriesekoop, F., Russell, C., Alvarez-Mayorga, B., Aidoo, K., Yuan, Q., Scannell, A., Beumer, R.R., Jiang, X. *et al.* (2010) Dirty money: an investigation into the hygiene status of some of the world's currencies as obtained from food outlets. *Foodborne Pathog Dis* **7**, 1497–1502.
- Watkinson, A., Murby, E., Kolpin, D.W. and Costanzo, S. (2009) The occurrence of antibiotics in an urban watershed: from wastewater to drinking water. *Sci Total Environ* **407**, 2711–2723.
- White, P.A., McIver, C.J. and Rawlinson, W.D. (2001) Integrons and gene cassettes in the enterobacteriaceae. *J Antimicrob Agents chemother* **45**, 2658–2661.
- Xiang, T., Chen, C., Wen, J., Liu, Y., Zhang, Q., Cheng, N., Wu, X. and Zhang, W. (2020) Resistance of *Klebsiella pneumoniae* strains carrying blaNDM-1 gene and the genetic environment of blaNDM-1. *Fron Microbiol* **11**, 110–120.
- Zhang, S.H., Lv, X., Han, B., Gu, X., Wang, P.F., Wang, C. and He, Z. (2015) Prevalence of antibiotic resistance genes in antibiotic-resistant *Escherichia coli* isolates in surface water of Taihu Lake Basin, China. *Environ Sci Pollut Res* **22**, 11412–11421.